

Temă - lab. 13

30 el. { group 1: [3, 5, 3, 2, 4, 6, 9, 3, 8, 10] $\rightarrow$ 10 el.
group 2: [10, 8, 15, 9, 11, 16, 17, 15, 7, 10] $\rightarrow$ 10 el.
group 3: [29, 15, 14, 15, 17, 10, 8, 11, 18, 19] $\rightarrow$ 10 el.

Ex. 1: Apply One-Way ANOVA and post-hoc analysis (the Turkey's method or the Scheffé approach) on data.

One-Way Analysis:

$H_0: \overbrace{\mu_1 = \mu_2 = \mu_3}^{\text{the means are equal}}$

$$\begin{aligned} \text{sum}_{gr.1} &= 3 + 5 + 3 + 2 + 4 + 6 + 9 + 3 + 8 + 10 \\ &= 53 \end{aligned}$$

$$\begin{aligned} \text{sum}_{gr.2} &= 10 + 8 + 15 + 9 + 11 + 16 + 17 + 15 + 7 + 10 \\ &= 118 \end{aligned}$$

$$\begin{aligned} \text{sum}_{gr.3} &= 29 + 15 + 14 + 15 + 17 + 10 + 8 + 11 + 18 + 19 \\ &= 156 \end{aligned}$$

The grand mean: $\bar{x}$

$$\begin{aligned} \bar{x} &= \frac{\sum x_i}{N} = \frac{\text{sum}_{gr.1} + \text{sum}_{gr.2} + \text{sum}_{gr.3}}{10 + 10 + 10} \\ &= \frac{53 + 118 + 156}{30} = \frac{327}{30} = \underline{\underline{10,9}} \end{aligned}$$

The Sum of Squares for the

treatment term: SS between

$$\begin{aligned} \text{SS}_{\text{between}} &= S_1^2 + S_2^2 + S_3^2 \text{ OR} \\ &= \sum n_j (\bar{x}_j - \bar{x})^2 \end{aligned}$$

$$\text{sum}_{\text{opt.1}} = 53 \Rightarrow \bar{X}_1 = \frac{53}{10} = 5,3$$

$$\text{sum}_{\text{opt.2}} = 118 \Rightarrow \bar{X}_2 = \frac{118}{10} = 11,8$$

$$\text{sum}_{\text{opt.3}} = 156 \Rightarrow \bar{X}_3 = \frac{156}{10} = 15,6$$

$$\begin{aligned} SS_{\text{between}} &= 10(5,3 - 10,9)^2 + \\ &\quad 10(11,8 - 10,9)^2 + \\ &\quad 10(15,6 - 10,9)^2 \\ &= 10 \cdot [(-5,6)^2 + (0,9)^2 + 4,7^2] \\ &= 10 \cdot (31,36 + 0,81 + 22,09) \\ &= \underline{\underline{542,6}} \end{aligned}$$

SS over or the SS within:

$$\begin{aligned} SS_{\text{within (opt.1)}} &= 3(3 - 5,3)^2 + (5 - 5,3)^2 + (2 - 5,3)^2 \\ &\quad + (4 - 5,3)^2 + (6 - 5,3)^2 + (9 - 5,3)^2 \\ &\quad + (8 - 5,3)^2 + (10 - 5,3)^2 \\ &= 3 \cdot (-2,3)^2 + (-0,3)^2 + (-3,3)^2 + (-1,3)^2 \\ &\quad + (0,7)^2 + (3,7)^2 + (2,7)^2 + (4,7)^2 \\ &= 15,87 + 0,09 + 10,89 + 1,69 + 0,49 + \\ &\quad + 13,69 + 7,29 + 22,09 \\ &= \underline{\underline{72,1}} \end{aligned}$$

$$\begin{aligned}
 SS_{\text{within}} (\text{qr. 2}) &= 2(10 - 11,8)^2 + (8 - 11,8)^2 + \\
 &+ 2(15 - 11,8)^2 + (9 - 11,8)^2 + (11 - 11,8)^2 + \\
 &+ (16 - 11,8)^2 + (17 - 11,8)^2 + (7 - 11,8)^2 \\
 &= 2(-1,8)^2 + (-3,8)^2 + 2(3,2)^2 + (-2,8)^2 + \\
 &+ (-0,8)^2 + (4,2)^2 + (5,2)^2 + (-4,8)^2 \\
 &= 6,48 + 14,44 + 20,48 + 7,84 + 0,64 + \\
 &+ 17,64 + 27,04 + 23,04 = \underline{117,56}
 \end{aligned}$$

$$\begin{aligned}
 SS_{\text{within}} (\text{qr. 3}) &= 2(15 - 15,6)^2 + (14 - 15,6)^2 + \\
 &+ (17 - 15,6)^2 + (19 - 15,6)^2 + (8 - 15,6)^2 + \\
 &+ (11 - 15,6)^2 + (18 - 15,6)^2 + (19 - 15,6)^2 \\
 &+ (29 - 15,6)^2 \\
 &= 2(-0,6)^2 + (-1,6)^2 + (1,4)^2 + (-5,6)^2 \\
 &+ (-7,6)^2 + (-4,6)^2 + (2,4)^2 + (3,4)^2 + (13,4)^2 \\
 &= 0,72 + 2,56 + 1,96 + 31,36 + 57,76 \\
 &+ 21,16 + 5,76 + 11,56 + 179,56 \\
 &= \underline{312,4}
 \end{aligned}$$

$$\begin{aligned}
 SS_{\text{within}} &= 72,1 + 117,56 + 312,4 \\
 &= \underline{502,06}
 \end{aligned}$$

The mean square between: MS_b

$$MS_b = \frac{SS_{\text{between}}}{k - 1}, \quad k - 1 = \text{the degrees of freedom} \\
 (3 \text{ qr.} - 1)$$

$$= \frac{542,6}{3 - 1} = \frac{542,6}{2} = \underline{271,3}$$

The mean square within (error): MS_w

$$\begin{aligned} MS_w &= \frac{SS_{\text{within}}}{N - k}, \quad df_{\text{error}} = df_w \\ &= df_{\text{error}} = 30 - 3 = 27 \\ &\quad (30 \text{ people} - 3 \text{ gr.}) \\ &= \frac{502,06}{27} \\ &= \underline{\underline{18,59}} \end{aligned}$$

F - statistic: F_{observed}

$$F_{\text{observed}} = \frac{MS_b}{MS_w} = \frac{271,3}{18,59} = \underline{\underline{14,59}}$$

	SS	df	MS	F_{obs}
Between	542,6	2	271,3	14,59
Within	502,06	27	18,59	
Total	1044,66	29		

$F_{\text{critical}}: \alpha = 0,05; df = 2; df_w = 27$

$$\Rightarrow F_{\text{critical}} = \underline{\underline{3,354}}$$

$F_{\text{observed}} > F_{\text{critical}} \quad (14,59 > 3,354)$

$\Rightarrow$ reject the null hypothesis
($\mu_0 = \mu_1 = \mu_2$)

Tukey and Scheffe post-hoc tests

The formula for the Tukey's HSD ^{threshold}

post-hoc test:
$$\frac{\bar{x}_1 - \bar{x}_2}{\sqrt{MS_w \left(\frac{1}{m_i} \right)}}$$

$n_1 = n_2 = n_3 = 10$

$\bar{x}_1 = 5,3 ; \bar{x}_2 = 11,8 ; \bar{x}_3 = 15,6$

$MS_w = 18,59$

Critical value for $k=3$: 3,51

Gr. 1 vs. Gr. 2	Gr. 1 vs Gr. 3	Gr. 2 vs Gr. 3
$\frac{ 5,3 - 11,8 }{\sqrt{\frac{18,59}{10}}}$	$\frac{ 5,3 - 15,6 }{\sqrt{\frac{18,59}{10}}}$	$\frac{ 11,8 - 15,6 }{\sqrt{\frac{18,59}{10}}}$
$\frac{6,5}{\sqrt{1,859}}$	$\frac{10,3}{\sqrt{1,859}}$	$\frac{3,8}{\sqrt{1,859}}$
$\frac{6,5}{1,363}$	$\frac{10,3}{1,363}$	$\frac{3,8}{1,363}$
$4,768 > 3,51$	$7,556 > 3,51$	$2,787 \leq 3,51$
$\Rightarrow$ reject the H_1 ($\bar{x}_1 = \bar{x}_2$)	$\Rightarrow$ rejects the H_2 ($\bar{x}_1 = \bar{x}_3$)	$\Rightarrow$ fails to reject H_3 ($\bar{x}_2 = \bar{x}_3$)

Minimum significant difference

for Scheffe test: $\sqrt{(k-1) \cdot F_{\text{critical}} \cdot MS_w \cdot \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$

$$= \sqrt{(3-1) \cdot 3,35 \cdot 18,59 \cdot \frac{2}{10}} = \sqrt{24,9106} = \underline{4,991}$$

Gr. 1 vs. Gr. 2	Gr. 1 vs Gr. 3	Gr. 2 vs Gr. 3
$ 5,3 - 11,8 $	$ 5,3 - 15,6 $	$ 11,8 - 15,6 $
6,5 > 4,991	10,3 > 4,991	3,8 < 4,991
$\Rightarrow$ Reject the H_1 ($\bar{x}_1 = \bar{x}_2$)	$\Rightarrow$ Reject the H_2 ($\bar{x}_1 = \bar{x}_3$)	$\Rightarrow$ fail to reject the H_3 ($\bar{x}_2 = \bar{x}_3$)

Scheffe Grouping	Mean	N	Group
A	15,6	10	3
A	11,8	10	2
B	5,3	10	1

For both Tukey and Scheffe

- rejected H_1 ($\bar{x}_1 = \bar{x}_2$)
- rejected H_2 ($\bar{x}_1 = \bar{x}_3$)
- fail to reject H_3 ($\bar{x}_2 = \bar{x}_3$)

Confidence Intervals:

$$t\text{-value} = n-1 = 9 \quad (\text{two-tailed test for } \alpha=0,05) \\ \leadsto 2,262$$

$$\text{Standard error} = SE = \sqrt{\frac{MSW}{n}} = \sqrt{\frac{18,59}{10}} = 1,3637$$

$$\begin{aligned} \text{Margin of error} = ME &= t \cdot SE \\ &= 2,262 \cdot 1,3637 \\ &= 3,0846 \end{aligned}$$

Ranges:

$$\begin{aligned} gr_1: \quad 5,3 \pm 3,0846 \\ \Rightarrow 2,2154 \longleftrightarrow 8,3846 \\ \\ gr_2: \quad 11,8 \pm 3,0846 \\ \Rightarrow 8,7154 \longleftrightarrow 14,8846 \\ \\ gr_3: \quad 15,6 \pm 3,0846 \\ \Rightarrow 12,5154 \longleftrightarrow 18,6846 \end{aligned}$$

Group	is it in interval?
1	$\bar{X}_2: 11,8$ is outside the range $\bar{X}_3: 15,6$ is outside the range
2	$\bar{X}_1: 5,3$ is outside the range $\bar{X}_3: 15,6$ is outside the range
3	$\bar{X}_1: 5,3$ is outside the range $\bar{X}_2: 11,8$ is outside the range